

Tutorial 37

Your paper, WRDS, the rules — and debt capacity by hand

Advanced Corporate Finance (EBM232A05)

University of Groningen

- **First hour — the project becomes concrete:** papers and parameter sheets are allocated; WRDS onboarding; the data rules that protect you.
- **Second hour — Problem set 37:** the debt-capacity model at the board.

Due next week (22:00 Tuesday): the **scoping memo** (D0) — your first graded deliverable, returned with the proceed verdict at assessment moment M1 in week 39.

Your paper and your parameter sheet

What you received today

- **Your paper** — allocated from your rankings, with a cap per paper so no one is over-subscribed.
- **Your parameter sheet** — unique to your team: (i) the **target sub-table / column** you must reproduce (6–10 target cells: coefficients, SEs, N , R^2), and (ii) the **extension axis** for D4.
- No two teams share a sheet. A generic AI answer, or another team's file, **does not fit yours** — and is detected at intake.

This week: read the paper **twice** — once for the argument, once for the recipe (sample, variable definitions, the exact table). The scoping memo is the recipe, written down.

The stack, and what to do

- **WRDS account** — register under the RUG subscription with your *@rug.nl* address. **Approval takes a day: do it today.**
- **Git + shared team repo** — one repo per team, from the released skeleton (TIER folders, *run_all* stub).
- **Python + the *wrds* package** (or SAS/Stata on the WRDS cloud) — the pull runs on **your** machine.
- **Le Chat / Vibe** on your RUG account — the AI that writes and debugs the pipeline with you.

The setup guide on Brightspace walks through every step; its closing checklist is this week's homework.

WRDS: first connection

Three libraries carry the course: **Compustat** (*comp.funda* — fundamentals), **CRSP** (prices and returns), and the **CCM link** that joins them.

```
import wrds
db = wrds.Connection() # prompts for WRDS username + password once
db.raw_sql("select gvkey, datadate, at from comp.funda limit 5")
```

- Five rows back = you are connected. Say **yes** to the *.pgpass* offer — credentials stored once, never typed into code.
- From the first pull on: **record every query and its pull date** in *data/raw-metadata*. Data vintage is a legitimate reconciliation category at M2 — *only* if documented.

The bright lines (they protect you)

- **AI writes the pipeline; you run the pull.** No cloud AI can, or may, connect to WRDS with your credentials.
- **Never** paste raw WRDS rows into any AI tool — licensed vendor data, regardless of model or hosting.
- **Never** commit raw extracts. Ship the **query specification** and the **derived dataset** — exactly how journals deposit replication packages.
- Your WRDS password lives in `.pgpass` — never in code, never in a commit.

Git from day one: **pull** → **work** → **commit small** → **push**, with messages that say what happened. The history is graded (D2b) — and it *defends* you in week 43.

Problem set 37, discussed

At $t = 0$ the firm chooses debt $D_0 \leq \bar{D}$, earning shield τ per euro. With probability p a project (cost I , NPV G) arrives at $t = 1$ and can be financed **only** by unused debt capacity: it needs $\bar{D} - D_0 \geq I$.

$$V(D_0) = \tau D_0 + pG \cdot \mathbf{1}\{D_0 \leq \bar{D} - I\} \quad \implies \quad \text{preserve} \iff pG \geq \tau I.$$

Board: only two candidates can be optimal (why?); the comparison; and the option reading — which side is the “dividend”, which the live option? Note $\bar{D}$ **drops out**.

$$\tau = 25\%, \bar{D} = 150, I = 50, p = 0.6, G = 30.$$

- Both candidate values: preserve = $0.25 \times 100 + 18 = \mathbf{43}$; lever = $0.25 \times 150 = 37.5$. **Preserves**, under-levering by $I = 50$; flexibility worth **5.5**.
- The inversions: indifference at $p^* = \tau I / G \approx \mathbf{0.417}$ and $\tau^* = pG / I = \mathbf{36\%}$ — both are **ratios of the two sides of the same inequality**.
- Higher $\tau \Rightarrow$ *less conservative* — and Week 36 says the relevant τ is the small, post-clawback one, which is why flexibility wins so often.

Now the shortfall *can* be covered by equity at cost κ per euro.

slope of V beyond $\bar{D} - I$: $\tau - p\kappa \implies \text{preserve} \iff \tau < p\kappa.$

- The cliff becomes a **kink**; the condition turns **marginal** — I and G drop out.
- $\kappa \rightarrow \infty$: the lecture model. $\kappa \rightarrow 0$: **MM** — flexibility is worthless when outside finance is free. **The friction doing the work is κ — Week 38's opening line.**

Frank and Goyal (2009): leverage ↓ in market-to-book and profitability, ↑ in size, tangibility, industry median.

- Market-to-book: flexibility *and* Myers (1977) overhang predict the sign — they **agree**, which is why it is the robust one.
- Tangibility moves $\bar{D}$ — the **level**, not the preserve/lever threshold ($\bar{D}$ cancelled in P1).
- Profitability: shared with the pecking order — **one matching sign is not a test**. The sign a *static* trade-off cannot deliver: market-to-book.
- Close: the Week-37 ledger row on the board.

Before you leave

- **Today:** register for WRDS; clone the team repo; first Vibe session.
- **This week:** paper read twice; variable-definition table drafted; test pull green.
- **Next Tuesday 22:00:** scoping memo (D0, 10 points, proceed gate). Feedback at M1, week 39.
- **Problem set 37 key** on Brightspace after today; the debt-capacity Explorer is the self-check.

- Frank, M. Z. and V. K. Goyal (2009). **“Capital Structure Decisions: Which Factors Are Reliably Important?”** In: *Financial Management* 38.1, pp. 1–37.
- Myers, S. C. (1977). **“Determinants of Corporate Borrowing”**. In: *Journal of Financial Economics* 5.2, pp. 147–175.